Potentials

Elias Xu

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1 Electrical Potentials

Electric potentials are measures of the potential energy created by an electric field. You can think of them as such:

Potential: Potential Energy :: Field: Force Energy

For the potential energy of point charges, there is a very big positional element here. As long as you have two charged particles, as long as they want to move somewhere, there will always be potential energy in that system. Usually, systems want to reduce that energy to zero.

1.1 Formulas and Dimensional Analysis

Let's look at the formulas for volts and field, and see what we can make for them. Let's have the formula for volts, and put of the formula for electric field for comparison.

$$V = \frac{U}{q} \qquad \vec{E} = \frac{\vec{F}}{q}$$

Let's do a quick dimensional analysis too for these *volts*:

$$[V] = \frac{[J]}{[C]}$$

Differences between field and potential:

- Field is a vector quantity, voltage is a scalar quantity

- Field lines are always perpendicular accross vector lines

Let's have a visualization changes of potential over time:

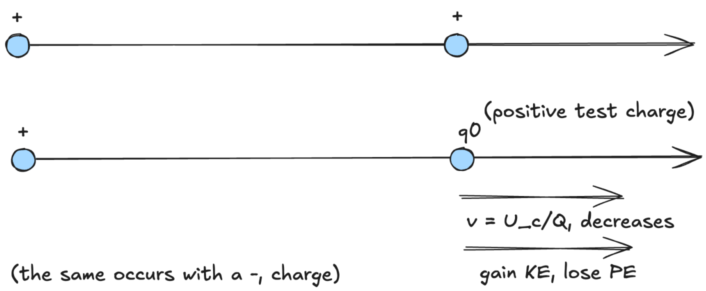

Figure 1: The change of charge accross a number line

Notes:

- Voltage decreases as one goes through a field lines

1.2 Equipotentials

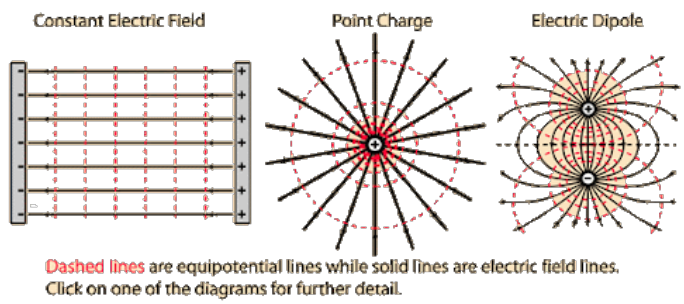

Figure 2: Example of electric fields vs electric potentials

- Lines accross these show the areas of equidistant potentials
- Circular around point charge
- *Perpendicular to field lines*, (e.g. Electric field is the negative gradient of potential), or $E = -\Delta V$

1.3 Calculating Electrical Potential from Electric Field

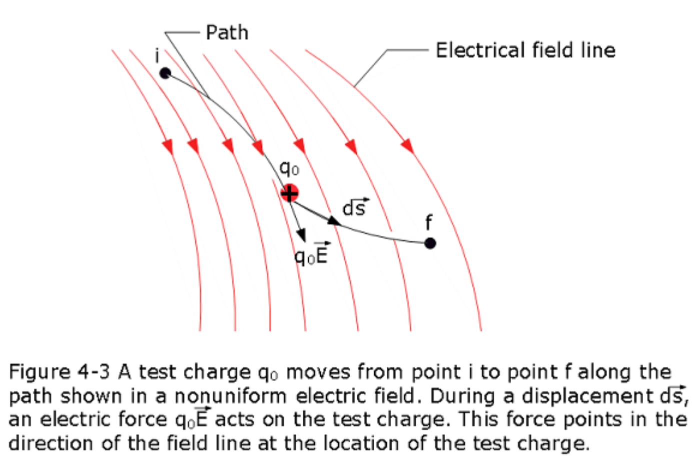

Figure 3: Visualization of the different differentials here

To try to find a formula for this, starting from the givens and working up.

$$\vec{F} = q_0\vec{E} \qquad\qquad W = \int \delta W$$

$$W = \int \delta w = \int_i^f \vec{F}\delta\vec{s}$$
$$= -\Delta U = -q_0\Delta V$$
$$\Delta v = \boxed{-\int_i^f \vec{E}\cdot\delta\vec{s}}$$

Given that electric field is a vector, and potentials are scalars, one might have to treat these as partial derivatives of x and y. So we have the original equation:

$$\vec{E} = \frac{Q}{4\pi\epsilon R^2} = -\frac{\delta V}{\delta s}$$

But we have to split it into components:

$$E_x = \frac{\delta V}{\delta x} \qquad\qquad E_y = \frac{\delta V}{\delta y}$$

$$\vec{E} = -\frac{\Delta V}{\Delta S} = \frac{\delta V}{\delta x} + \frac{\delta V}{\delta y}$$

2 The Electic Potential of a Point Charge

Given a ring of radius R, which has a uniformly distributed charge Q, we can infer the following:

- The net electric field in the center is zero (common sense)
- The net electric field at every point is also zero (similar to gravatational field)
- What is the potential inside the ring, given $\Delta V = \int \vec{E}\cdot\delta\vec{s} = \int 0\cdot\delta\vec{s} = \boxed{C}$.

2.1 Calculating the Point Charge's Potential Through Integration

Things to note:

- Potential is a function of charge and radius, or the distance from a point
- Charge creates a scalar potential field
- $u(q_1, q_0) = q_0 V(q_1)$, the relationship between charge and potential energy

To get definite zero, we can set the potential at $r \to \infty$ to be zero, or $\lim_{r\to\infty} V(q) = 0$ to be zero.

$$\Delta V = V_f - V_i = 0 - V_i = \int_r^\infty \vec{E}\cdot\delta\vec{s}$$
$$-V_r = -\int_r^\infty \frac{1}{4\pi\epsilon_0}\frac{Q}{r^2}[\hat{r}\cdot\delta\vec{s}] = -\int_r^\infty \frac{1}{4\pi\epsilon_0}\frac{Q}{r^2}\delta r$$
$$\boxed{V_r = \frac{Q}{4\pi\epsilon r}}$$

Note that the property is *scalar*. This makes stuff like addition and subtraction better!

3 Electric Flux

A measure of how many field lines pierces a certain surface / goes through a surface

3.1 Finding formula for Electric Flux

Flux is porportional to

- Cross sectional area of the surface [A]
- Strengh of the wires [E]
- orientation of the wires [θ]

Max flux occurs when $\theta = 0°$ (the angle between the normal of the surface and the field lines), and mix flux occurs when $\theta = 90°$ (e.g. when normal of the surface and the field lines are perpendicular).

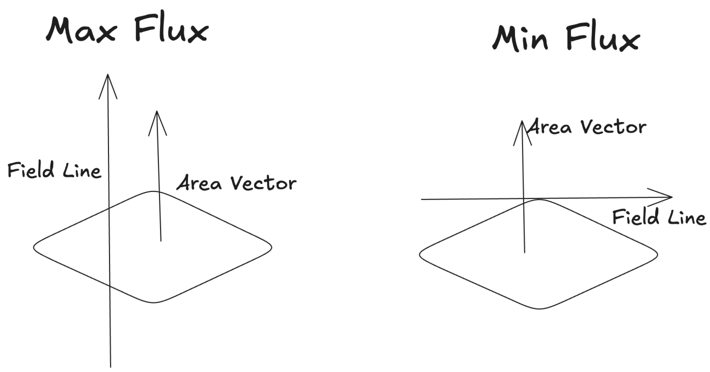

Figure 4: Visualization of max and min flux

Thus, you can compute the flux here:

$$\Phi = \vec{E} \cdot \vec{A} = |\vec{E}||\vec{A}| \cos\theta$$

This is with the assumption that that

- Plane is regular
- Field is constant

For an irregular plane / field, you can use this formula

$$\Phi = \int \vec{E} \cdot \delta\vec{A}$$

3.2 Closed Areas vs. Open Areas

Gaussian Surfaces are closed figures, and that allows you to find a direction for the area vector, which is always outwards. Some additional points about close areas and flux:

- The flux for an object that is hit by continious equipotentials and doesn't contain any of their own is zero
- For example, if you have flux that goes through a box, the flux is zero

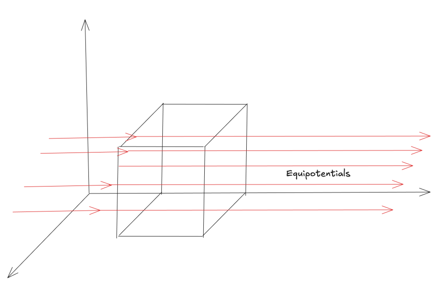

Figure 5: Object with zero flux

4 Gauss's Law

Eletric flux gives us **Gauss' Law**, which tells us that the eletric flux through a closed surface is porportional to the enclosed charge.

4.1 Gauss' Theorem

Gauss' theorem relates flux to charge, field, and area:

$$\Phi = \frac{q_{\text{enc}}}{\epsilon_0} = \int \vec{E} \cdot \delta \vec{A}$$

This can also be related to potentials, through the calc idea of divergence (iirc)

$$\Delta \cdot E = \frac{\rho}{\epsilon_0}$$

These equations, though, are only good/useful if you have a good gaussian surface

4.2 Nice Gaussian Surfaces

An example is if it is symmetrical to a charged object. An example is a circular surface with a point charge in the center.

$$\Phi = \frac{q_{\text{enc}}}{\epsilon_0} = \int \vec{E} \cdot \delta \vec{A}$$
$$= E \int \delta A = \frac{Q}{\epsilon_0}$$
$$E(4\pi r^2) = \frac{Q}{\epsilon_0}$$
$$E = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2}$$

(Interesting because this merges both empeirical physics with pure mathematics)

4.3 Electrical Fields vs Flux

Eletric fields and flux can vary, for but flux you can calculate a total since it's a scalar value

5 Charges on a Conductor

Any changes placed in an insulated conductor would go the edges of the condutor, none will remain.

5.1 Proof by Gauss' law

Let's solve with a copper (pretty good condutor) and inside it there's a gaussian surface.

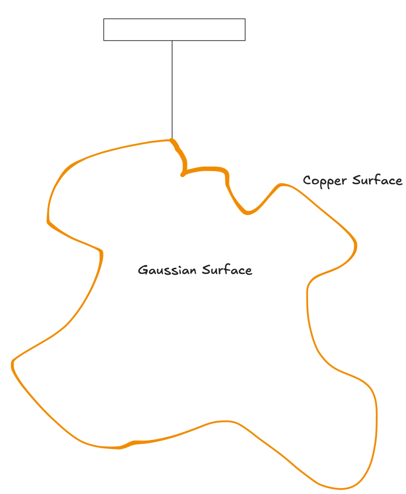

Figure 6: Gaussian setup

We have to prove that the field everywhere except the surface is zero, which means that $E_{\text{Gaussian surface}} = 0$, because then, through Gauss' theorem, the charge would then be zero. Let's prove that through negation:

$$E \neq 0 \to \exists E \to \exists F_{\text{on any q}} \to \exists a_{\text{on any q}}$$

An acceleration means that there definitely is no field.

5.2 Charge inside a cavity

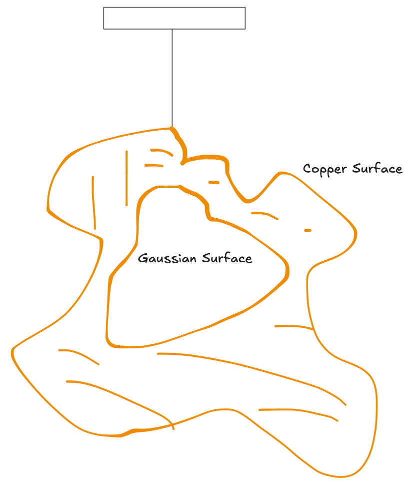

Figure 7: Gaussian setup

The implications of this is that if there was a charge inside a cavity, it would be balanced out by a charge along the wall of the conductor to make sure that the charge is zero. If the charge inside the sirface was q, then the charge on the edge would be $-q$.